Spatial Revising Variational Autoencoder-Based Feature Extraction Method for Hyperspectral Images

Wenbo Yu, *Member, IEEE*, Miao Zhang, *Member, IEEE*, and Yi Shen, *Member, IEEE*

***Abstract*— Hyperspectral image with high dimensionality always increases the computational consumption, which challenges image processing. Deep learning models have achieved extraordinary success in various image processing domains, which are effective to improve classification performance. There remain considerable challenges in fully extracting abundant spectral information, such as the combination of spatial and spectral information. In this article, a novel unsupervised hyperspectral feature extraction architecture based on spatial revising variational autoencoder (AE) (U_{Hfe}SRVAE) is proposed. The core concept of this method is extracting spatial features via designed networks from multiple aspects for the revision of the obtained spectral features. Multilayer encoder extracts spectral features, and then, latent space vectors are generated from the obtained means and standard deviations. Spatial features based on local sensing and sequential sensing are extracted using multilayer convolutional neural networks and long short-term memory networks, respectively, which can revise the obtained mean vectors. Besides, the proposed loss function guarantees the consistency of the probability distributions of various latent spatial features, which obtained from the same neighbor region. Several experiments are conducted on three publicly available hyperspectral data sets, and the experimental results show that U_{Hfe}SRVAE achieves better classification results compared with comparison methods. The combination of spatial feature extraction models and deep AE models is designed based on the unique characteristics of hyperspectral images, which contributes to the performance of this method.**

***Index Terms*— Convolutional neural network (CNN), deep learning, feature extraction, hyperspectral image, variational autoencoder (VAE).**

I. Introduction

HYPERSPECTRAL imagery has hundreds of narrow and adjacent spectral bands and provides valuable information about the ground surface of the earth [1]. Studies show that hyperspectral data actually reside on a low-dimensional manifold. Due to this unique intrinsic geometric structure, hyperspectral data are widely used in the field of target recognition, land-cover classification [2], [3], and urban planning [4]. Considering that hyperspectral data contain a large amount of bands, proposing novel feature extraction methods becomes extremely urgent to dispose of these problems [5], [6].

The aim of feature extraction methods is to transform the raw data into a specific latent space with lower dimension, preserving enough discriminate information for identifying samples [7]. Based on types of transformation frameworks, feature extraction methods can mainly be categorized into linear models [8] and nonlinear ones. Existing linear techniques, consisting of models such as principal components analysis (PCA) [9] and linear discriminant analysis (LDA) [10], perform well when dealing with linear data sets. For the sake of investigating local manifold properties of nonlinear data sets, they are extended into nonlinear ones based on kernel function, such as kernel LDA and kernel ICA [11]. Moreover, manifold learning methods, such as Laplacian eigenmap [12], local linear embedding [13], and local tangent space alignment [14], are prominent in complex contexts. For real data sets, these traditional but nontrivial methods laid foundations for future research.

In the era of acknowledge explosion, traditional models are faced with dilemmas when analyzing large-scale data sets. Deep learning techniques generally become mainstream when coping with specific tasks in the machine learning domain [15], [16]. In feature extraction tasks, convolutional neural networks (CNNs) [17]–[19] and autoencoders (AEs) stand out in different practical fields. The former is famous for its sparse connectivity and weight sharing [20]. The latter targets samples rather than complete images, which is suitable for real hyperspectral data. In these decades, a variety of improved AE-based neural network architectures are proposed for complex contexts, such as denoising AE [21], sparse AE [22], [23], and variational AE [24]–[26]. Moreover, recurrent neural network [27], [28] and its variants, such as long short-term memory (LSTM) network [29]–[32], are used to model sequential data and extract the relationship features from them.

As promising options to address difficulties, deep learning techniques train architectures by iterations and parameter optimization [33]. Gradient descent finds an optimal direction to reduce the amount of multivariable loss function [34].

Manuscript received June 27, 2019; revised November 10, 2019 and January 10, 2020; accepted April 22, 2020. Date of publication June 5, 2020; date of current version January 21, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61876054 and in part by the Postdoctoral Scientific Research Developmental Fund of Heilongjiang Province under Grant LBH-Q17078. *(Corresponding author: Miao Zhang.)*

The authors are with the Department of Control Science and Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: ieeeywb@126.com; ieeemiao@126.com; yishen_hit@126.com).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2997835

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

Common gradient descent optimization algorithms include Adam, RMSprop, and Adagrad. Network architectures are excellently trained by proposing suitable loss functions and choosing appropriate gradient descent optimization algorithms.

Researchers have introduced deep learning techniques into hyperspectral imaging and proposed multiple DNN methods for hyperspectral feature extraction [16], [35]. Some variants of AE are widely used in remote sensing domain [36], [37]. Scientists fully utilized spectral information, and methods proposed work well when combining spectral information with AE-based networks. Zabalza *et al*. [38] proposed a novel segmented stacked AE (S-SAE) for hyperspectral feature extraction, which confronted the original features into small parts to reduce the complexity. An unsupervised nonlinear stacked DAE was presented by Ahmad *et al*. [39], which also confronted original data into small parts. VAE makes the latent feature follow a normal distribution by appending constraints to the encoder. Although these approaches achieve great classification performance, they do not fully take advantage of the abundant spatial information.

Similar to other high-resolution multispectral data such as the Worldview-2 images, hyperspectral data contains multiple spatial characteristics, so it is inappropriate to ignore them [40]–[42]. More and more network models are proposed to extract joint hyperspectral spectral–spatial features in an end-to-end manner. Belwalkar *et al*. [43] proposed a spectral–spatial classification model using VAE and CNN. Zhou *et al.* [32] proposed a feature extraction method based on spectral–spatial LSTM networks. It extracted spectral–spatial features without taking the data restore into consideration. Chen *et al.* [44] proposed a stacked AE-based spectral–spatial DNN technique (SAELR) merging these features using SAE. Chen *et al.* [33] also presented a stacked CNN for hyperspectral spectral–spatial feature extraction, which tries to address the imbalance between high dimensionality and limited availability of training samples.

Since the positions of pixels are interrelated in the spatial domain [45], the local perspective of one single pixel can be regarded as sequential data [32]. In this article, a novel feature extraction method based on spatial revising variational AE is proposed. This method combines multiple network architectures commendably and mainly contains four modules: data preprocessing and sample preparation, spectral feature encoding, spatial feature encoding (local sensing and sequential sensing), and decoding. The first module generates local sequential data sets based on their spatial locations. The spectral encoding module finds optimal latent features of the original single-pixel vector via a multilayer network. Needed means when obtaining latent space vectors are generated from spectral features via VAE, which are revised by obtained spatial features via stacked LSTM and stacked CNN, respectively. The spatial revising strategy is specially designed for this generative model, which amplifies the advantage of VAE in this domain. Stacked networks extract spatial features from two aspects, making the revised means better and making the obtained latent space vectors comprehensive for classification. A novel loss function is proposed by introducing the Kullback–Leibler (KL) divergence to guarantee the consistency of probability distributions of various latent spatial features, which are obtained from the same neighbor region. This article contributes to the literature in three major respects.

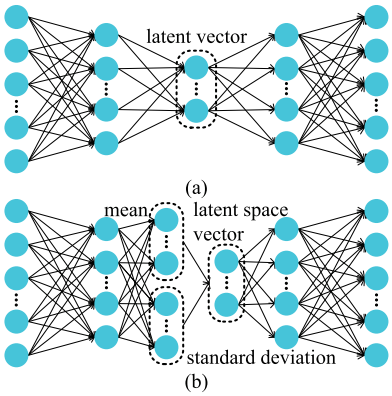


Fig. 1. Architectures of standard (a) AE and (b) VAE.

1) A novel deep learning-based hyperspectral feature extraction method is proposed to extract the spectral–spatial feature representation by constructing a well-designed architecture. Considering that the positions of pixels in hyperspectral data sets are interrelated, we represent the spectral–spatial information via sequential data rather than feature vectors.
2) Spatial features are extracted via stacked LSTM (sequential sensing) and stacked CNN (local sensing) from two different aspects. Then, they are utilized to revise the obtained spectral features. It should be noted that the revision is adaptive to achieve a balanced state of multispatial features. To the best of our knowledge, this is the first time that VAE has been used for unsupervised hyperspectral feature extraction.
3) To ensure the homology of latent spatial features obtained from the same neighbor region, the proposed loss function for this DNN architecture introduces KL divergence. It minimizes the difference between input data and generated data. In the meantime, the proposed function guarantees the homology of multiple latent features in the spatial field and ensures the consistency of their probability distributions.

The remainder of this article is organized as follows. After introducing the related works in Section II. Section III presents the proposed method in detail, including its structure and explanation. Experimental results on classification are reported in Section IV, followed by the conclusion in Section V.

II. Related Works

In this section, some key related works on involved networks are presented comprehensively.

A. Variational Autoencoders

Variational AE (VAE) is one variant of the standard AE [24] (architectures of them are shown in Fig. 1). The essence of VAE is to add a "Gaussian noise" to the result of encoders, which makes the result obtained after decoding more robust

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

to data noise. The latent embeddings of hyperspectral images always reside in low-dimensional space. Taking account of the complexity of real high-dimensional data, VAE has advantages over AE considering of noise interference [26]. For one sample x (one-pixel vector), VAE assumes that the expected sample $\hat{x}$ (has high similarity with x) can be generated from the latent space vector t. The latent space vector t is generated from $p_\alpha(t)$ (prior distribution), and the expected sample is generated from $p_\alpha(x|t)$ (conditional distribution); α shows the parameters in the decoder network. Considering that $p_\alpha(t|x)$ (posterior distribution) is intractable, Kingma and Welling [24] proposed using $q_\beta(t|x)$ (Gaussian distribution) as a substitute for $p_\alpha(t|x)$, and β shows the parameters in the encoder network. The mean μ and the standard deviation σ can be obtained using multilayer. The KL divergence is used to measure the distance between $p_\alpha(t|x)$ and $q_\beta(t|x)$. The evidence lower bound (ELBO) function is proposed in [26], [36] as

$$\text{ELBO} = \mathbb{E}_{q_\beta(t|x)} p_\alpha(x|t) - KL[p_\alpha(t|x), q_\beta(t|x)] \tag{1}$$

where $E_{q_\beta(t|x)} p_\alpha(t|z)$ is the generation probability of x, which can be calculated using cross entropy or mean squared error (MSE). Considering that $p_\alpha(t|x)$ cannot be calculated, Kingma and Welling [24] regarded it as $\mathcal{N}(0, 1)$. Then, we get

$$\mathcal{N}(\mu, \sigma^2) = \mu + \mathcal{N}(0, 1) \times \sigma. \tag{2}$$

Therefore, the second part in (1) is rewritten as

$$\begin{aligned} KL[p_\alpha(t|x), q_\beta(t|x)] &= KL[\mathcal{N}(0, 1), \mathcal{N}(\mu, \sigma^2)] \\ &= \frac{1}{2}(-\log\sigma^2 + \mu^2 + \sigma^2 - 1). \end{aligned} \tag{3}$$

The mean can be considered as "pure latent feature" generated from original data, and the standard deviation is the added "Gaussian noise." The latter is used to dynamically adjust the noise intensity. When the fitting process is unfinished, the noise will be decreased appropriately, which makes it easier to converge. If the network becomes stable, it will be increased to improve the generation ability of the whole network.

This model has the ability of resisting noise interference, which is beneficial to hyperspectral data processing. As a typical generative model, VAE generates the latent space features from spectral information, and it is necessary to use both mean and standard deviation, which ensures that the whole network is robust and will not degenerate into a classical AE [24], [26]. It will become more suitable for hyperspectral images after introducing spatial information.

B. LSTM Networks and Convolutional Neural Networks

LSTM and CNN extract spatial features from hyperspectral images, and they play key roles in different aspects. LSTM was proposed by Hochreiter and Shmidhuber [29] for sequential data, and then, it was extended by Graves [30]. The architecture of LSTM is shown in Fig. 2, and $\otimes$ and $\oplus$ are scalar product and addition for matrices, respectively. The whole structure consists of three gates: forget gate, input gate, and output gate. As can be seen in Fig. 2, the process of LSTM

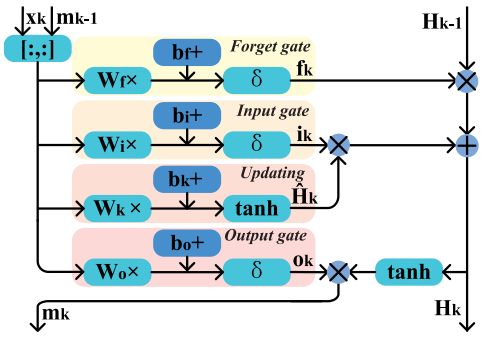


Fig. 2. Architectures of LSTM.

(x_k is the kth sample in the sequential data, m_{k-1} is the previous output, and H_{k-1} is the previous state of model) [32] is shown as

$$\begin{aligned} f_k &= \delta(W_f \times [x_k, m_{k-1}] + b_f) \\ i_k &= \delta(W_i \times [x_k, m_{k-1}] + b_i) \\ \hat{H}_k &= \tanh(W_k \times [x_k, m_{k-1}] + b_k) \\ o_k &= \delta(W_o \times [x_k, m_{k-1}] + b_o) \\ H_k &= (f_k \otimes H_{k-1}) \oplus (i_k \otimes \hat{H}_k) \\ m_k &= \tanh(H_k) \otimes o_k \end{aligned} \tag{4}$$

where δ is the activation function. (W_f, b_f), (W_i, b_i), and (W_o, b_o) are parameters (weights and bias) for the three gates, and (W_k, b_k) are parameters for model updating.

Compared with LSTM designed for data continuity, CNN takes neighbor regions surrounding target vectors in original data as inputs, fully considering spatial information of the whole image [18], [19]. Multichannel convolutions work well in extracting spatial features as well as reducing spectral bands. Feature maps are obtained via multilayer convolution, and the 2-D operator at each layer [19] is as follows:

$$x_{cd}^{yz} = \hat{\delta}\left(b_{cd} + \sum_n \sum_{u=0}^{U_c-1} \sum_{v=0}^{V_d-1} W_{cdn}^{uv} x_{(c-1)n}^{(y+u)(z+v)}\right) \tag{5}$$

where x_{cd}^{yz} is a neuron at position (y, z) of the dth feature map in the cth layer, $\hat{\delta}$ is the activation function, b_{cd} is the bias of the dth feature map in the cth layer, n indexes feature maps connected to the cth layer in the $(c-1)$th layer, W_{cdn}^{uv} is the weight of position (u, v) connected to the n feature map, and U_c and V_c are sizes of convolution kernels (the height and width).

These networks are outstanding in multiple aspects when extracting spatial features, which can be used to revise spectral features. Therefore, in this article, a novel spectral–spatial DNN architecture is proposed to fully utilize various kinds of features in hyperspectral images.

III. Methodology

In this section, an unsupervised spectral–spatial feature extraction method named U_{Hfe}SRVAE for hyperspectral images based on spatial revising variational AE is proposed. Its flowchart is shown in Fig. 3, where $A@B$ means that totally, B samples exist, and each sample has A features. This method consists of four parts: data preprocessing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

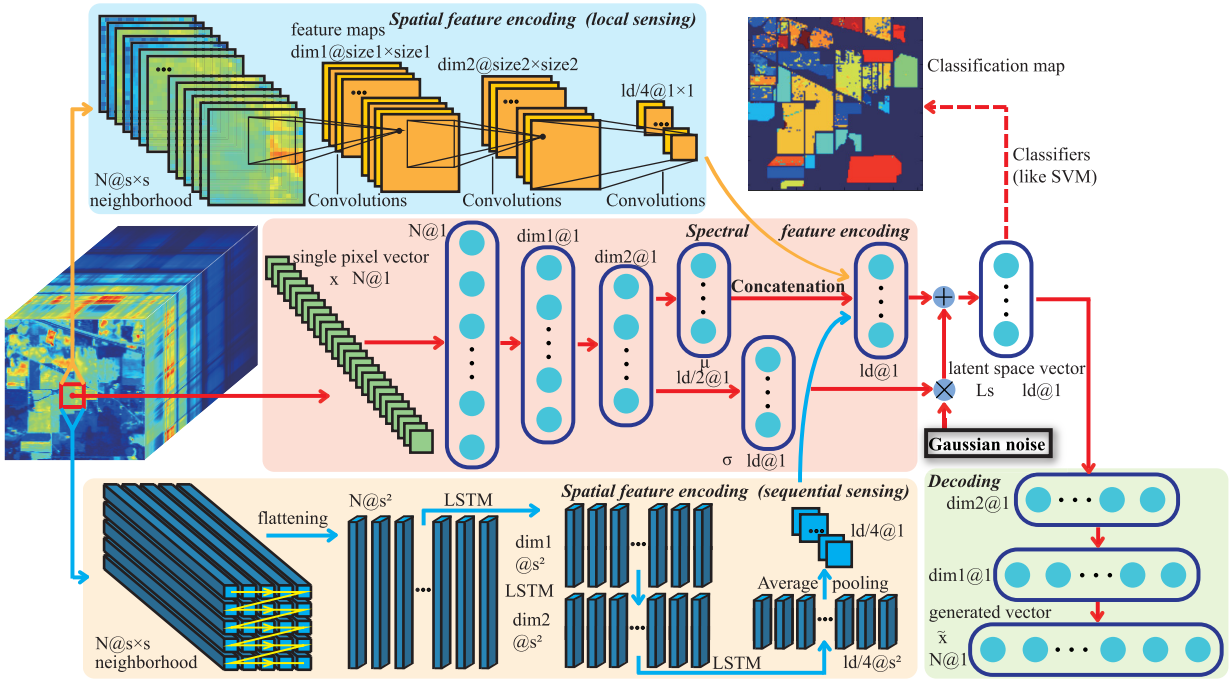


Fig. 3. Flowchart of the proposed U_{Hfe}SRVAE (the number of stacked layers in encoding/decoding parts can be changed).

and sample preparation, spectral encoding, spatial encoding (including local sensing and sequential sensing), and decoding.

A. Data Preprocessing and Sample Preparation

The hyperspectral image is denoted as $\mathbf{X} \in \mathbb{R}^{l\times w\times N}$, which comprises the $l \times w$ pixels and has N spectral reflectance bands. Considering that VAE is an independent sample processing network model, pixel vectors are fed into stacked dense layers after normalizing/standardizing the whole hyperspectral image. Pixel vectors contain spectral features rather than spatial features. Neighbor regions surrounding pixel vectors comprise local spatial features, which is utilized in different aspects.

LSTM can process sequential data and extract relationships among them. Positions of pixels in the spatial domain are fixed and interrelated, so one-pixel vector $x \in \mathbb{R}^{1\times 1\times N}$ and its neighbor region sorted in a specific order can be considered as sequential data. The neighbor region is chosen by a $s \times s$ neighboring window, containing s^2 pixel vectors. The general sorting method is right-to-left and top-to-bottom, and the number of sequential samples equals to that of pixels. For pixels on the edge of images, a mirror symmetry method is used (see Fig. 4). CNN also takes neighbor regions of pixels as inputs. Different from LSTM, it mainly recognizes the local spatial distribution of whole images, so sorting samples is unnecessary.

B. Spectral Feature Encoding

Due to the high dimensionality of hyperspectral images, a tremendous amount of spectral information exists. Lights with different wavelengths are reflected on objects, conveying different spectral characteristics. However, too many channels lead to difficulties in identifying samples, and it always results

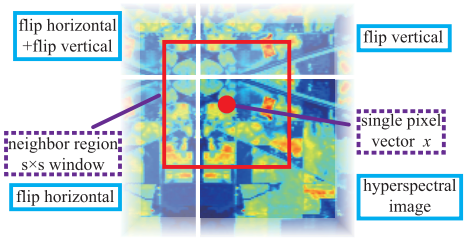


Fig. 4. Mirror symmetry method in U_{Hfe}SRVAE.

in a waste of time. Extracting discriminant features seems to be significant to tackle these problems. The facts show that DNN architectures extract increasingly significant features at high layers (too many layers may lead to overfitting). AE is suitable for hyperspectral data; however, it cannot deal with inevitable noise compared with VAE.

In the proposed method, VAE with stacked dense layers extracts the spectral features from original data. Each pixel vector $x \in \mathbb{R}^{1\times N}$ in the hyperspectral image is one independent sample, where N is the number of spectral bands. Samples are fed into stacked dense layers, and then, the mean μ and standard deviation σ can be obtained. It is noticeable that μ and σ should have the same size. Since μ only contains spectral features, it is revised by spatial features in Section III-C. Parameters in these layers are trained using the proposed loss function in Section III-E. The mean μ is taken for latent "pure spectral feature," and the benefit of σ^2 is to add a "Gaussian noise" to μ, which increases its robustness to noise.

C. Spatial Feature Encoding

Spatial features are extracted from multiple aspects to revise obtained μ, and the number of layers in both parts is the same as that of spectral feature encoding for consistency.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED SIZES OF CONVOLUTION KERNELS

index	kernel size	size of result	number of filters
1	$\lfloor s/2 \rfloor + 1$	size_1	dim_1
2	$\lfloor \text{size}_1/2 \rfloor + 1$	size_2	dim_2
...	...	...	...
ρ-1	$\lfloor \text{size}_{\rho-2}/2 \rfloor + 1$	$\text{size}_{\rho-1}$	$\text{dim}_{\rho-1}$
ρ	$\text{size}_{\rho-1}$	1	ld (latent dimension)

1) Local Sensing: Spatial distribution in different bands contains abundant information [45]. Local sensing utilized in stacked CNN means that spatial features are extracted from receptive fields rather than whole data. For one-pixel vector x, its neighbor region contains s^2 pixel vectors. As shown in Fig. 3, feature maps are generated by selecting different sizes of convolution kernels. Considering that the aim of extracting spatial features is to revise μ, the size of the last feature map x_{ls} should equal to that of μ, and sizes of convolution kernels are shown in Table I (suppose that there are ρ convolutional layers), where $\lfloor \bullet \rfloor$ is the rounding down operator, the kernel size and the size of the result are sizes of one side of the convolution kernel and obtained result, respectively. Three convolutional layers are shown in the flowchart, and if more or fewer layers are used, sizes of convolution kernels should be changed accordingly to control the size of the last feature map.

2) Sequential Sensing: Sequential sensing means extracting spatial features from sequential data. Pixel vectors in hyperspectral images are irrelevant in the time domain; however, positions of them are interrelated in the spatial domain. The whole image will lose its spatial characteristics if these positions of pixels are disturbed, so the neighborhood region of one single pixel vector can be regarded as sequential data. Similar to the previous part, the neighbor region of one-pixel vector x is chosen by a $s \times s$ window, it is flattened and elements are sorted into $\{x_1, x_2, \ldots, x_{s^2}\}$ according to the method in Section III-A. The time step is s^2, and these sorted elements are fed into stacked LSTM to extract spatial features. The input vectors are interrelated, and the target pixel x resides in the center of the neighborhood region, so the obtained result is fed into an average pooling layer to generate the final result x_{ss}.

3) Spatial Revision: Both kinds of spatial feature vectors $x_{ls} \in \mathbb{R}^{1\times 1\times ld/4}$ (adapted from $\mathbb{R}^{1\times 1\times ld/4}$) and $x_{ss} \in \mathbb{R}^{1\times ld/4}$ are used to revise the mean $\mu \in \mathbb{R}^{1\times ld/2}$, where ld is the expected dimension of the latent vector. They reflect various aspects of the whole data. Dimensions of them are $ld/4$, $ld/4$, and $ld/2$ for guaranteeing the final dimension after concatenation. They are concatenated into a complete latent mean, and then, the obtained mean is fed into the whole network architecture. Moreover, there should be a relationship between the spatial features extracted from the same neighbor region or at least the probability distributions of them should be consistent with each other. Considering that, we introduce the KL divergence into the proposed loss function (see Section III-E), which assists with narrowing the spatial distance between spatial directions of various latent spatial features. The KL divergence would not make them identical but guarantee their homology.

D. Decoding

After revising the "pure latent feature" and obtaining the complete latent mean, "Gaussian noise" based on the obtained standard deviation is obtained, and we get

$$L_s = \text{Concatenation}_{[x_{ls}, x_{ss}, \mu]} \oplus (\mathcal{N}(0, 1) \otimes \sigma). \tag{6}$$

$\text{Concatenation}_{[x_{ls}, x_{ss}, \mu]}$ concatenates three kinds of features, and their dimensions match with each other. The latent space vector L_s is fed into stacked dense layers that have symmetrical numbers of nodes with those in the spectral feature encoding module, and its aim is to recover the input.

E. Loss Function

The aim of standard AE is to reconstruct input data and minimize the reconstruction error between the input data and the generated data. VAE constructs its loss function (1) by introducing KL divergence. A novel loss function $\Gamma = \Gamma_{\text{R}} + \Gamma_{\text{KL}} + \Gamma_{\text{homology}}$ for this method is proposed, which consists of three parts. MSE-based Γ_{R} calculates the similarity between the input $x \in \mathbb{R}^{1\times N}$ and the output $\tilde{x}$, which is formulated as

$$\Gamma_{\text{R}} = \sum_{j}^{N} (x_j - \tilde{x}_j)^2 \tag{7}$$

where N is the number of spectral bands. Similar to the second part in (1), Γ_{KL} is formulated as

$$\Gamma_{\text{KL}} = (-\log \sigma^2 + (\text{Concatenation}_{[x_{ls}, x_{ss}, \mu]})^2 + \sigma^2 - 1)/2. \tag{8}$$

Γ_{homology} guarantees the homology of two different spatial features, x_{ls} and x_{ss}. As shown in Fig. 3, local sensing and sequential sensing extract spatial features utilizing the same neighbor region. In other words, they represent characteristics from various perspective, but they should be homologous (not the same but have similar probability distributions). This homology is achieved by introducing the KL divergence as

$$\Gamma_{\text{homology}} = \frac{1}{2} \sum_{j}^{ld} \left(x_{ls\,j} \times \log \frac{x_{ls\,j}}{x_{ss\,j}} + x_{ss\,j} \times \log \frac{x_{ss\,j}}{x_{ls\,j}} \right). \tag{9}$$

Considering that inputs of KL divergence are not interchangeable, two parts are used to change this characteristic.

F. Architecture Optimization

For DNN architectures, network optimization algorithms are effective for parameter optimization. Classically, weights and bias for all layers should be determined by iterations. For example, these parameters exist in intermediate variables (such as $\tilde{x}$, x_{ls}, and x_{ss}) in the proposed loss function Γ.

1) Gradient Descent: Parameters share similar optimization processes, so they are all represented as θ for brevity. Based on the Lagrange mean value theorem, $\Gamma(\theta_t)$ is expanded as

$$\Gamma(\theta_t) = \Gamma(\theta_{t-1}) + (\theta_t - \theta_{t-1}) \bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1}) \tag{10}$$

where the partial derivative $\bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1})$ is defined as

$$\bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1}) = d\Gamma(\theta_{t-1})/d\theta_{t-1}. \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

Then, the optimized θ_t is obtained as

$$\theta_t = \theta_{t-1} - \eta \bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1}) \tag{12}$$

where θ_{t-1} is the same parameter before the tth iteration, and η is the learning rate.

2) Adaptive Moment Estimation: Improved optimization algorithms have been proposed, such as the adaptive moment estimation (Adam) [46], the adaptive gradient algorithm (AdaGrad) [47], and the root mean square prop (RMSprop) [48]. The Adam, proposed by Kingma [46], is introduced in this article because of its better optimization performance. It combines fundamental theories and advantages of AdaGrad and RMSprop [46]

$$\begin{aligned} c_t &= \beta_1 \times c_{t-1} + (1-\beta_1) \bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1}) \times \bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1}) \\ d_t &= \beta_2 \times d_{t-1} + (1-\beta_2) \bigtriangledown_{\theta_{t-1}} \Gamma(\theta_{t-1}) \\ \hat{c}_t &= \frac{c_t}{1-\beta_1}, \quad \hat{d}_t = \frac{d_t}{1-\beta_2}, \quad \theta_{t+1} = \theta_t - \frac{\eta}{\hat{c}_t + \varepsilon} \times \hat{d}_t. \end{aligned} \tag{13}$$

Parameters in Adam under Keras framework have the default setting as $\beta_1 = 0.999$, $\beta_2 = 0.9$, and $\varepsilon = 0$ (or set as a small value like 10^{-8}). Other detailed information can be seen in [46].

IV. Experimental Results and Discussion

Several experiments are conducted on three hyperspectral data sets,[1] which are widely used in remote sensing field, to prove the effectiveness of the proposed U_{Hfe}SRVAE.

A. Data Description

The first data set (Indian Pines data set) was acquired by the AVIRIS sensor in Northwest Indiana in 1992. It has 145×145 pixels and 220 spectral bands in the 0.4–2.5-μm wavelength, and the spatial resolution is 20 m. Twenty bands were removed because of the water absorption. The second data set (PaviaU data set) was acquired over the area of Pavia University in 2002. Its spatial resolution is 1.3 m. It consists of 610×340 pixels with 115 bands in the wavelength range of 0.43–0.86 μm, in which 12 noisy bands were removed. The third data set (Salinas data set) was collected by the AVIRIS sensor over Salinas Valley, which consists of 512×217 pixels with 224 bands. Twenty bands were discarded because of water absorption and noise. Tables II–IV show their ground-truth classes, and Figs. 5–7 show these data sets at length.

B. Experimental Settings

The purpose of this technique is to extract the spectral–spatial features by neural network architectures, so some machine/DNN-based techniques are included for benchmarking, including LLE [13], AE, segmented stacked AE (S-SAE) [38], and variational AE (VAE). They are pixelwise methods without using spatial neighbors. LLE is a typical nonlinear manifold feature extraction method considering local manifold structure of samples [13]. All the samples are used in LLE since its methodology makes it difficult to extract features using limited samples. AE is a basic DNN architecture for hyperspectral data. S-SAE confronts the original hyperspectral data into smaller data segments and processes

[1]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

TABLE II
Detailed Total/Training Samples in the Indian Pines Data Set

Label	Land cover type	Samples	Training
1	Alfalfa	54	6
2	Corn-No Till	1434	143
3	Corn-Min Till	834	84
4	Corn	234	23
5	Grass-Pasture	497	50
6	Grass-Trees	747	74
7	Grass-Pasture-Mowed	26	3
8	Hay-Windrowed	489	49
9	Oats	20	4
10	Soybean-No Till	968	97
11	Soybean-Min Till	2468	247
12	Soybean-Clean	614	61
13	Wheat	212	22
14	Woods	1294	129
15	Buildings-Grass-Trees-Drives	380	38
16	Stone-Steel-Towers	95	10

TABLE III
Detailed Total/Training Samples in the PaviaU Data Set

Label	Land cover type	Samples	Training
1	Asphalt	6631	664
2	Meadows	18649	1864
3	Gravel	2099	210
4	Trees	3064	307
5	Painted Metal Sheets	1345	134
6	Bare Soil	5029	503
7	Bitumen	1330	133
8	Self-Blocking Bricks	3682	368
9	Shadows	947	95

TABLE IV
Detailed Total/Training Samples in the Salinas Data Set

Label	Land cover type	Samples	Training
1	Brocoli-Green-Weeds-1	2009	201
2	Brocoli-Green-Weeds-2	3726	373
3	Fallow	1976	198
4	Fallow-Rough-Plow	1394	139
5	Fallow-Smooth	2678	268
6	Stubble	3959	396
7	Celery	3579	358
8	Grapes-Untrained	11271	1127
9	Soil-Vinyard-Develop	6203	620
10	Corn-Senesced-Green-Weeds	3278	328
11	Lettuce-Romaine-4wk	1068	107
12	Lettuce-Romaine-5wk	1927	192
13	Lettuce-Romaine-6wk	916	92
14	Lettuce-Romaine-7wk	1070	107
15	Vinyard-Untrained	7268	727
16	Vinyard-Vertical-Trellis	1807	180

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

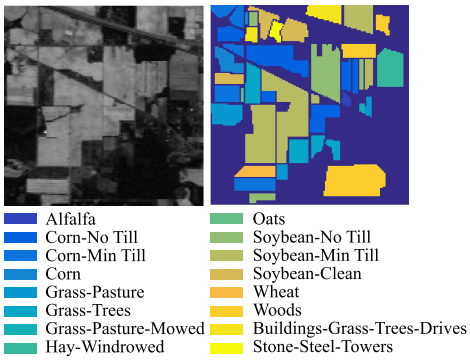


Fig. 5. Visualization of the Indian Pines data. (Left) Gray image of the 150th band. (Right) Ground truth.

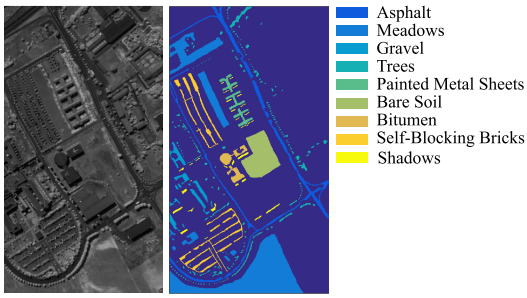


Fig. 6. Visualization of the PaviaU data. (Left) Gray image of the 100th band. (Right) Ground truth.

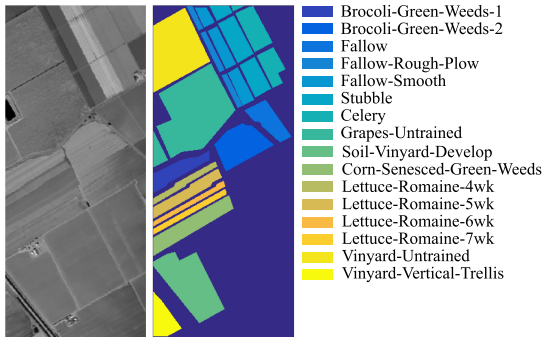


Fig. 7. Visualization of the Salinas data. (Left) Gray image of the 100th band. (Right) Ground truth.

them by different smaller SAEs separately [38]. For AE and VAE, parameters used are the same with those in the proposed method.

Moreover, two typical spectral–spatial DNN methods are also chosen for comparison. The first one is an encoder–decoder-based DNN method (SAELR) [44], which is a spectral–spatial feature extraction method utilizing stacked AE. The second one is a conventional NN-based architecture (S-CNN) [33], which extracts the spectral–spatial features using stacked CNN. They are both supervised methods, so the fine-tuning of architectures of them is carried out by introducing decoders (convert to unsupervised methods) for a fair comparison. Other parameters are set based on their articles [33], [44].

For DNN methods, 30% samples are randomly selected as the training set to construct networks. To ensure the integrity of classification results, all samples are utilized to test the obtained DNN models. Batch normalization is used to keep inputs of different layers having identical distribution. The batch size is set as 512, the Adam algorithm is selected as the optimizer, the learning rate is set as 0.001, and the best results are recorded within 10^3 iterations for a fair comparison.

The support vector machine (SVM) using LIBSVM [7] is applied as the classifier to evaluate the obtained results; 10% samples in each class are randomly selected for training in classification (20% samples in the ninth class are used in the Indian Pines data set), and the remaining samples are used for testing. The detailed numbers are listed in Section IV-A. Fivefold cross validation is used to choose optimal parameters, and necessary parameters (penalty coefficient and gamma in RBF kernel) are selected from a grid search within the same set $\{2^{-4}, 2^{-3}, \ldots, 2^4\}$. Overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are obtained by the SVMs to evaluate different methods. κ is an index that evaluates classification accuracies. Suppose that totally g classes exist, z_i is the number of samples within class i in the ground-truth map, and $\hat{z}_i$ is the number of samples within class i in the classification map. κ is defined as

$$\kappa = \frac{\mathrm{OA} - \mathrm{Acc}}{1 - \mathrm{Acc}} \tag{14}$$

where OA is the overall classification, Acc is a constructed accuracy defined as $(z_1 \times \hat{z}_1 + z_2 \times \hat{z}_2 + \cdots + z_g \times \hat{z}_g)/(M \times M)$, and M is the number of whole samples.

C. Hyperparameter Analysis

Hyperparameters play an important role in the construction of DNN and they could affect classification results to some extent. Significant hyperparameters in U_{Hfe}SRVAE, including the number of hidden nodes, the number of hidden layers, and the size s of neighbor region, should be determined first. Considering that selecting different numbers of nodes for all hidden layers is time-consuming and unnecessary, we make all previously unmentioned hidden layers have the same number of nodes, which is equal to desired latent dimensions (ld).

1) Layer Depth Analysis: Experiments are conducted on the Indian Pines data set to show the effect of layer depth on classification results. We set s as 11 and select the optimal number of layers (spectral/spatial feature encoding and decoding share the same number of layers for consistency), and the results evaluated by three indexes are shown in Table V.

It shows that the OAs for most dimensions increase initially as the network deeper, and then, they drop slightly when more than three layers are used. When ld is 40, using three layers increases OA by 2.80% than that of using five layers, and the increase is 3.95% when ld is 8. There are two reasons for the results. First, although deepening architectures correspond to high-level features generally, it does not work all the time. For the Indian Pines data set with 10 366 samples, networks with too many hidden layers are hard to be trained, which may lead to the decrease of classification appearance. Second, this setting is not just a question

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

TABLE V

CLASSIFICATION RESULTS OA (%), AA (%), AND κ (%) OBTAINED BY DIFFERENT LAYER DEPTHS (SIZE s IS SET AS 11, AND THE SHAPE OF NEIGHBOR REGIONS IS 11×11)

Latent Dimension	1 layer			2 layers			3 layers			4 layers			5 layers		
(ld)	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
8	67.47	56.51	62.25	80.85	74.41	78.04	80.91	70.14	78.11	78.59	70.98	75.43	76.96	67.69	73.49
16	77.29	69.30	73.93	83.42	80.88	81.05	87.90	84.10	86.21	82.45	77.62	79.90	82.02	72.84	79.41
24	80.30	76.02	77.39	83.24	78.21	80.82	86.79	86.54	84.91	85.77	82.47	83.72	86.76	80.63	84.86
32	82.64	77.03	80.13	84.39	76.38	82.15	89.30	86.08	87.78	86.84	83.05	84.94	83.81	77.14	81.49
40	85.51	82.11	83.44	86.81	81.35	84.92	91.39	87.49	90.18	87.83	85.37	86.11	88.51	84.62	86.87

TABLE VI

CLASSIFICATION RESULTS OA (%), AA (%), AND κ (%) OBTAINED BY DIFFERENT WINDOW SIZES s (NUMBER OF LAYERS IS SET AS 3)

Latent Dimension	s=3			s=5			s=7			s=9			s=11		
(ld)	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
8	77.59	68.81	74.24	75.22	67.42	71.42	75.72	68.18	72.03	77.80	66.57	74.52	80.91	70.14	78.11
16	81.78	72.51	79.12	84.42	75.16	82.15	84.88	83.11	82.70	85.05	76.74	82.87	87.90	84.10	86.21
24	83.34	74.68	80.91	84.17	80.44	81.89	84.49	83.00	82.28	86.36	80.48	84.43	86.79	86.54	84.91
32	82.33	77.70	79.81	85.21	81.37	83.08	85.65	80.58	83.58	88.60	86.03	86.98	89.30	86.08	87.78
40	84.58	75.56	82.35	85.78	80.76	83.73	86.92	80.45	85.05	88.58	81.46	86.96	91.39	87.49	90.18

of deepening the network in these experiments. As shown in Table I, the sizes of convolution kernels in local sensing are related to the number of hidden layers. For example, when $s = 11$ and three hidden layers exist, sizes of convolution kernels are [6, 4, 3]. If four/five layers exist, sizes are [6, 4, 2, 2]/[6, 4, 2, 2, 1]. More hidden layers correspond to wider neighbor regions, so the depths should be carefully chosen if better results are expected. When three layers exist in spectral/spatial feature extraction parts, great classification appearances are obtained for all indexes. Therefore, setting this hyperparameter as three is suitable.

2) Neighbor Region Size Analysis: Three hidden layers are used for each part, and then, we select the optical size s from a given size {3, 5, 7, 9, 11} (too large size may lead to time-consuming). OA, AA, and κ are also utilized to evaluate the classification results with different values of s. Results in Table VI show that networks with different sizes of neighbor regions generate great results. As s increases, all indexes increase slightly since more spatial information is delivered into this network. However, the needed time for each epoch also rapidly extending. Times for each epoch ($ld = 40$) are 5.25, 14.27, 28.36, 48.93, and 74.65 s when $s = 3$, 5, 7, 9, and 11, respectively. Considering efficiency and accuracy, using too bigger s is needless. For the comprehensiveness of the study, we set s as 7 and 11 for all spectral–spatial techniques in our experiments.

D. Performance Comparison

In this section, experiments are conducted on three publicly available hyperspectral data sets to illustrate the superiority of the proposed U_{Hfe}SRVAE. For all hyperspectral images, features are extracted by different techniques, and then, they are classified by the same SVM classifier. All experiments are carried out using Python 3.7.0 (Spyder) and MATLAB R2016a on a PC with Intel Core i7-6700 CPU and 8-GB RAM.

1) Indian Pines Data Set: Experiments on the Indian Pines data set are conducted first. As shown in Fig. 3, the latent dimension should be the multiple of four, so we set it as 8, 16, 24, 32, and 40. For spectral–spatial methods, $s = 7$ and $s = 11$ are both chosen for comprehensiveness. The variations of OAs, AAs, and κs with respect to the latent dimensions are shown in Fig. 8. It shows that the classification results increase with the increase of latent dimension, and curves become smooth and slow generally. As a typical machine learning method, LLE has weakness compared with DNN methods. DNN spectral–spatial methods generally yield better results than pixelwise methods (AE, S-SAE, and VAE) because more spatial information is considered when training models. For DNN spectral–spatial techniques, classification results of U_{Hfe}SRVAE are higher than others with the same s on the whole, which means that the proposed technique outperforms other comparison methods. The OA for $U_{\text{Hfe}}\text{SRVAE}_{11}$ reaches 91.39% when the dimension is 40, which is 2.33% and 4.71% bigger than those of SAELR_{11} and S-CNN_{11}. Moreover, this method also increases accuracies by using spatial features, and its accuracy is 14.67% bigger than that of VAE, which is a pixelwise method. Considering that S-CNN and U_{Hfe}SRVAE both contain convolution layers, their computational efficiencies are compared with each other. Times for each epoch ($ld = 40$) are 15.76, 53.89, 28.36, and 74.65 s for S-CNN_7, S-CNN_{11}, $U_{\text{Hfe}}\text{SRVAE}_7$, and $U_{\text{Hfe}}\text{SRVAE}_{11}$, respectively. Although the proposed method needs more time to train the whole architecture, this investment is worth considering its great classification performance. As shown in (14), κ and OA are positively correlated, so their curves share similar patterns; however, the detailed values also show the superiority of the proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

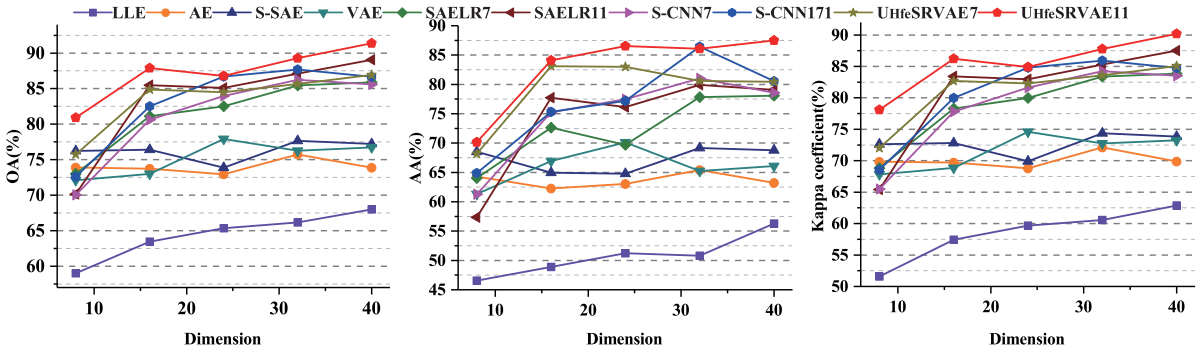


Fig. 8. Curves of classification accuracies (OA, AA, and kappa coefficient κ, respectively) on the Indian Pines data set with the number of dimensions increasing from 5 to 50.

TABLE VII

CLASSIFICATION ACCURACIES (%) OF COMPARISON METHODS FOR THE INDIAN PINES DATA SET, BOLD VALUES INDICATE THE BEST RESULTS FOR EACH ROW

Label	Original	LLE	AE	S-SAE	VAE	$SAELR_7$	$SAELR_{11}$	$S\text{-}CNN_7$	$S\text{-}CNN_{11}$	$U_{Hfe}SRVAE_7$	$U_{Hfe}SRVAE_{11}$
1	33.33	0.00	12.50	56.25	4.17	25.00	39.58	25.00	56.25	60.42	**89.58**
2	77.54	45.24	64.76	71.80	70.95	77.85	85.59	77.85	82.57	83.35	**89.39**
3	59.47	46.00	52.40	55.73	56.27	81.20	82.27	74.93	72.27	76.00	**85.07**
4	50.71	15.64	47.87	48.34	60.19	63.98	69.19	68.72	73.93	74.41	**81.99**
5	82.55	78.52	76.96	88.14	72.93	89.93	89.04	88.14	79.19	85.68	**92.62**
6	91.68	94.21	92.87	90.49	92.57	95.10	**98.22**	93.76	95.99	94.35	96.73
7	13.04	**65.22**	30.43	26.09	17.39	34.78	21.74	34.78	17.39	8.70	34.78
8	95.91	90.68	97.50	97.50	99.77	99.09	**100.00**	97.27	97.05	99.32	98.64
9	43.75	0.00	31.25	37.50	56.25	75.00	43.75	81.25	81.25	87.50	**93.75**
10	59.13	61.54	63.15	68.89	74.51	79.56	82.66	80.48	79.68	82.32	**89.90**
11	84.51	85.10	84.02	83.07	82.85	89.91	92.17	92.08	90.18	90.36	**93.16**
12	68.17	23.69	49.37	54.79	58.41	78.12	78.66	72.51	80.65	81.19	**85.53**
13	91.05	91.05	95.26	93.16	88.95	96.32	98.42	97.37	98.42	**99.47**	98.95
14	**98.28**	94.68	91.59	95.45	92.36	95.54	97.08	94.16	97.42	96.22	96.74
15	47.95	21.64	40.35	51.75	46.49	73.68	**89.18**	78.95	87.72	69.01	80.12
16	91.76	87.06	81.18	81.18	83.53	94.12	97.65	**100.00**	98.82	98.82	92.94
OA(%)	78.46	68.0	73.88	77.21	76.72	85.88	89.06	85.59	86.68	86.92	**91.39**
AA(%)	68.05	56.27	63.22	68.76	66.10	78.07	79.08	78.58	80.55	80.45	**87.49**
κ(%)	75.21	62.87	69.89	73.85	73.25	83.85	87.51	83.52	84.76	85.05	**90.18**

Then, we select one specific latent dimension (we set it as 40 for example) and show the great classification appearance of the proposed method by comparing the classification accuracies of classes. Table VII shows the classification accuracies acquired by comparison methods when the latent dimension is equal to 40. The "original" column means that all 200 bands are utilized. Compared with the results of the original data set, feature extraction improves the discriminative ability of samples. Although LLE does not work as well as other methods, samples in class 7 are classified better than other methods. Spectral–spatial feature extraction methods yield better classification results than pixelwise methods (AE, S-SAE, and VAE) since the latter can only extract spectral features. DNN spectral–spatial architectures work well, and the proposed U_{Hfe}SRVAE has the highest classification results. More concretely, samples in Classes 1–4, 5, and 9–12 are classified better than those of other techniques. This significant improvement indicates the superiority of the proposed method in extracting spectral–spatial features. Its variational part reduces noise interference and improve the robustness the whole structure by constructing the mean and the standard deviation. Fig. 9 shows full classification maps achieved by different methods on this data set when the reduced dimension is equal to 50. Some typical classes and parts are circled by red lines. For example, some samples in the 14th classes (Woods) are shown in the square in the lower right corner, we can see that they are correctly classified using the proposed method; however, their classification results using other methods such as SAELR and S-CNN are not as great as that of U_{Hfe}SRVAE. It can be observed that U_{Hfe}SRVAE obtains a more accurate map with less wrongly classified samples.

The 3-D visualization of data describes the performance of classification results and feature extraction architectures. The latent embedding clusters of this data set by deep network techniques and their three-view drawing are shown

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

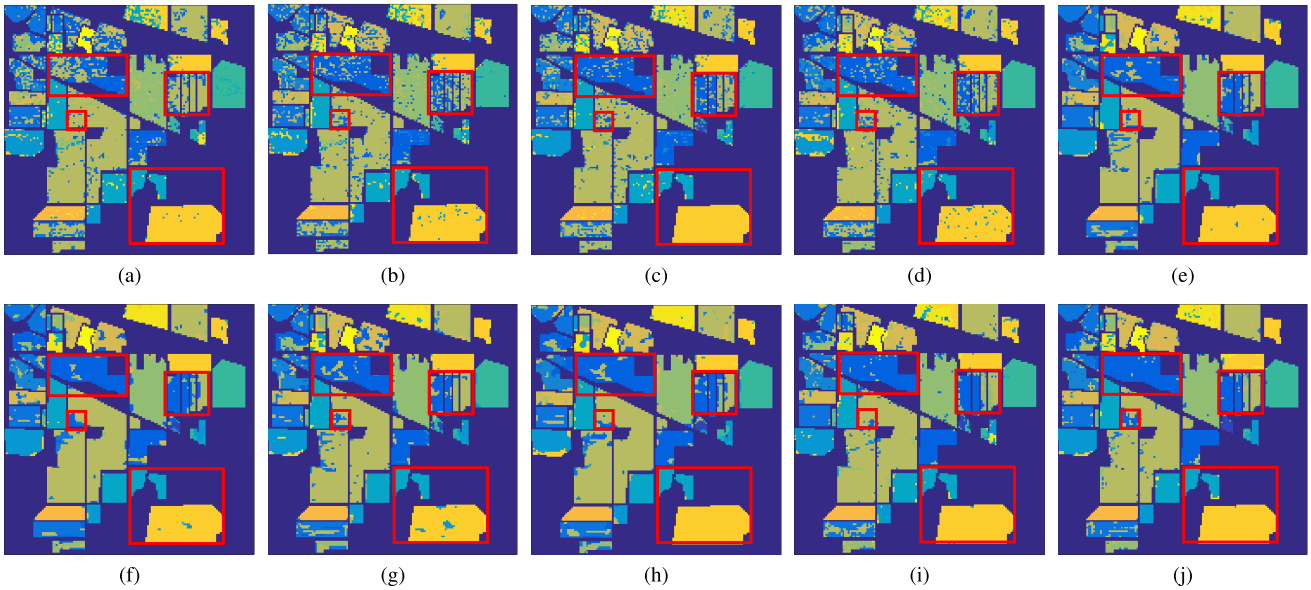


Fig. 9. Classification maps of the Indian Pines data set obtained by comparison methods. (a) LLE. (b) AE. (c) S-SAE. (d) VAE. (e) SAELR_7. (f) SAELR_{11}. (g) S-CNN_7. (h) S-CNN_{11}. (i) $U_{\text{Hfe}}\text{SRVAE}_7$. (j) $U_{\text{Hfe}}\text{SRVAE}_{11}$.

TABLE VIII

CLASSIFICATION ACCURACIES (%) OF USING MULTIPLE SPATIAL FEATURES FOR THE INDIAN PINES DATA SET, BOLD VALUES INDICATE THE BEST RESULTS FOR EACH COLUMN

ld		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	OA	AA	κ
16	7/lc	27.1	72.6	63.1	57.3	82.1	90.2	**65.2**	97.5	25.0	71.4	88.6	47.6	94.7	**97.1**	41.5	84.7	78.8	69.1	75.5
	7/sq	31.3	79.2	**76.5**	**68.7**	71.6	92.6	56.5	96.1	56.3	**79.6**	**90.8**	**74.5**	95.8	92.6	57.9	94.1	83.7	75.9	81.3
	7/all	**81.3**	**80.0**	76.0	60.2	**82.3**	**96.0**	52.2	**98.0**	**100.0**	78.0	88.5	73.1	**99.5**	95.4	**70.8**	**98.8**	**84.9**	**83.1**	**82.7**
	11/lc	14.6	75.4	74.9	71.1	70.7	92.6	**78.3**	98.6	62.5	78.9	83.3	65.1	97.4	87.8	51.5	96.5	80.0	75.0	77.1
	11/sq	**77.1**	84.6	**77.3**	63.5	86.4	92.6	73.9	96.6	75.0	80.3	**90.0**	63.5	96.3	91.5	53.2	**100.0**	84.4	81.3	82.2
	11/all	75.0	**85.5**	76.5	**80.1**	**91.1**	**97.3**	30.4	**98.9**	**100.0**	**82.3**	88.7	**85.0**	**99.5**	**95.7**	**77.2**	82.4	**87.9**	**84.1**	**86.2**
40	7/lc	47.9	76.7	72.1	72.0	89.0	92.9	39.1	98.0	75.0	77.8	88.8	75.0	98.9	92.4	62.9	83.5	83.6	77.6	81.2
	7/sq	**70.8**	81.3	74.7	62.6	**92.6**	**96.0**	**52.2**	98.2	68.8	75.5	90.3	74.3	98.9	**96.9**	**69.0**	**98.8**	85.8	**81.3**	83.8
	7/all	60.4	**83.4**	**76.0**	**74.4**	85.7	94.4	8.7	**99.3**	**87.5**	**82.3**	**90.4**	**81.2**	**99.5**	96.2	**69.0**	**98.8**	**86.9**	80.5	**85.1**
	11/lc	68.8	77.9	73.7	75.4	91.1	93.8	39.1	94.3	68.8	78.1	88.2	73.8	**99.5**	95.9	64.9	88.2	84.4	79.5	82.2
	11/sq	50.0	79.5	68.9	68.2	82.1	92.9	**43.5**	**98.6**	87.5	80.6	89.1	72.7	**99.5**	87.1	65.8	**94.1**	83.1	78.8	80.7
	11/all	**89.6**	**89.4**	**85.1**	**82.0**	**92.6**	**96.7**	34.8	**98.6**	**93.8**	**89.9**	**93.2**	**85.5**	99.0	**96.7**	**80.1**	92.9	**91.4**	**87.5**	**90.2**

in Fig. 10. They show great clusterability and discernibility. Fig. 10(a)–(c) shows a bit confusing since they are pixelwise methods. Fig. 10(h)–(i) shows that the proposed DNN architectures can discriminant samples from other classes accurately and then separate them, which also reflects that the proposed method has the best clustering performance.

Table VIII shows the necessity of using both kinds of spatial features (via local sensing and sequential sensing) and it makes the obtained vectors more comprehensive for classification. In this experiment, we set the latent dimension d as 16 and 40. In Table VIII, 7 and 11 mean $s = 7$ and $s = 11$, respectively, *lc* is short for "using local sensing part only," and *sq* means only using sequential sensing part. This experiment illustrates the effectiveness of using both kinds of spatial features and the necessity of fusing these spatial features and spectral features into low-dimensional features. For 7/all, samples in class 7 are not well classified, which is lower than 7/lc and 7/sq, and it results in a slight reduction in average accuracy. However, considering these three classification indices comprehensively, 7/all can achieve a better result. Then, their computational efficiencies are analyzed and compared with each other. Times for each epoch ($ld = 40$) are 5.77, 16.59, and 28.36 s for 7/lc, 7/sq, and 7/all, respectively. They are 19.70, 41.39, and 74.65 s for 11/lc, 11/sq, and 11/all, respectively. Although using various kinds of spatial features increase the running time, it is worthy of doing this. It can be seen that the running time of the proposed method is mostly allocated to LSTM because LSTM is harder to be trained than CNN. However, our aim is to increase the classification results rather than significantly reduce training time. Compared with the greater results than others, the slightly longer running time should not be a problem.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

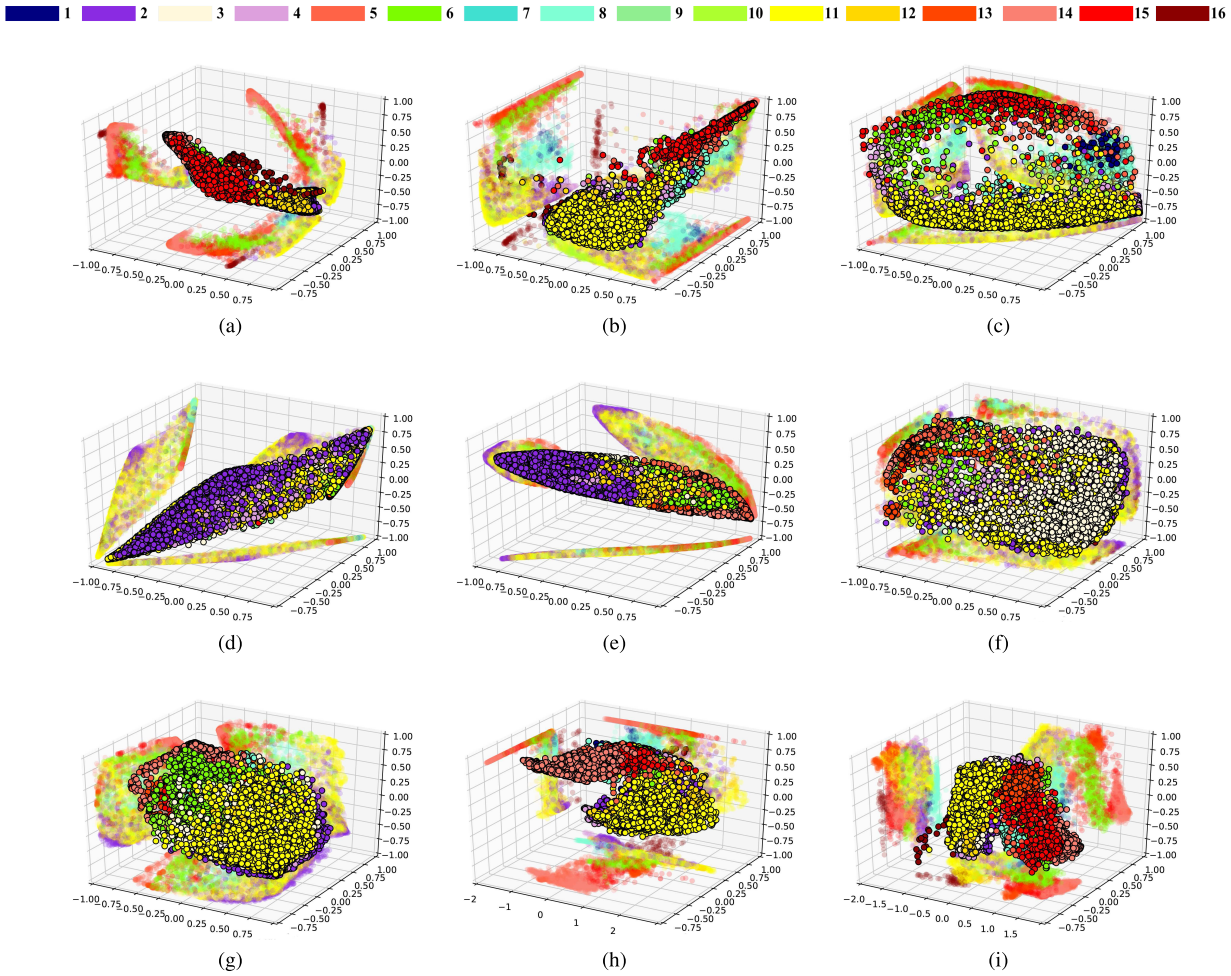


Fig. 10. Latent embedding through 3-D visualization of the Indian Pines data set obtained by the comparison methods. (a) AE. (b) S-SAE. (c) VAE. (d) SAELR$_7$. (e) SAELR$_{11}$. (f) S-CNN$_7$. (g) S-CNN$_{11}$. (h) U_{Hfe}SRVAE$_7$. (i) U_{Hfe}SRVAE$_{11}$.

TABLE IX

CLASSIFICATION ACCURACIES (%) OF THE COMPARISON METHODS FOR THE PAVIAU DATA SET, BOLD VALUES INDICATE THE BEST RESULTS FOR EACH ROW

Label	Original	LLE	AE	S-SAE	VAE	SAELR	S-CNN	U$_{\text{Hfe}}$SRVAE$^3_{lc}$	U$_{\text{Hfe}}$SRVAE$^3_{sq}$	U$_{\text{Hfe}}$SRVAE$_3$
1	92.54	88.99	87.45	92.31	92.09	95.19	94.00	93.63	95.19	**95.84**
2	**97.59**	96.69	96.50	97.56	97.18	97.28	96.98	96.63	95.99	97.24
3	72.31	38.43	61.73	75.49	70.04	75.22	79.78	73.32	79.04	**84.86**
4	88.54	81.25	86.04	89.52	87.12	**93.18**	92.49	90.75	87.41	93.04
5	98.43	98.68	99.09	98.93	99.42	**99.92**	**99.92**	**99.92**	99.50	**99.92**
6	81.26	40.50	64.60	85.04	80.87	75.85	69.89	79.85	81.60	**86.28**
7	75.77	67.84	77.19	85.13	83.21	84.96	80.62	81.62	83.88	**86.72**
8	83.98	84.58	86.30	88.99	89.68	93.15	92.52	92.40	92.88	**94.51**
9	89.79	**99.88**	99.65	99.77	99.41	99.41	99.77	99.77	99.30	99.77
OA	91.00	83.12	87.57	92.58	91.46	92.45	91.43	91.97	92.27	**94.40**
AA	86.69	77.43	84.28	90.30	88.78	90.46	89.55	89.76	90.53	**93.13**
κ	87.97	76.92	83.27	90.11	88.59	89.90	88.50	89.29	89.69	**92.55**

2) PaviaU Data Set: Then experiments are conducted on the PaviaU data set, and Table IX shows the classification accuracies acquired by the comparison methods when the latent dimension is equal to 40. For this data set, we only set s as 3 since classification performances when using other values of s have been shown in previous experiments, so there is no need to repeat it. Moreover, using the same s is fair for all spectral–spatial DNN methods. The "original" column means that all 103 bands are all utilized for classification. Although some samples of comparison methods are classified better than

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

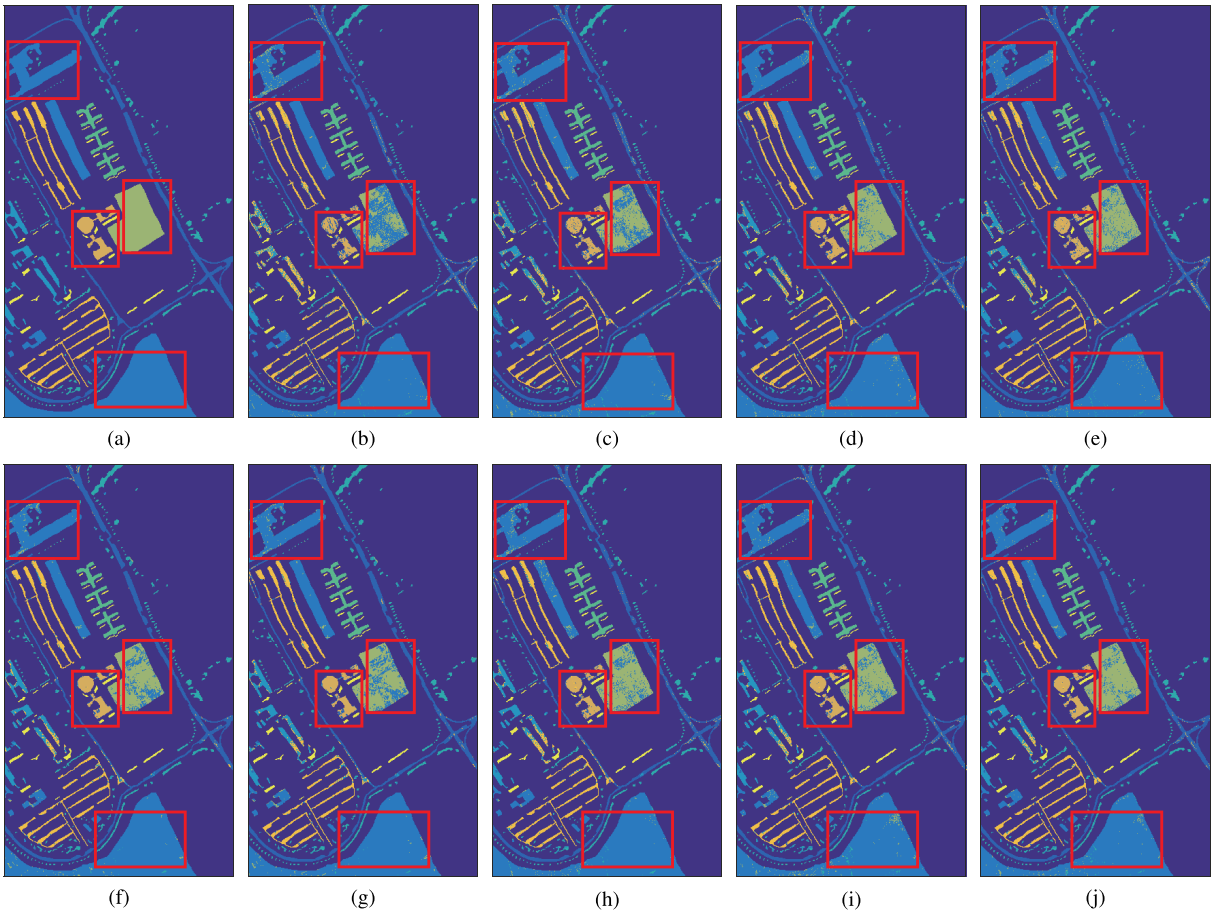


Fig. 11. Classification maps of the PaviaU data set obtained by the comparison methods. (a) Ground truth. (b) LLE. (c) AE. (d) S-SAE. (e) VAE. (f) SAELR. (g) S-CNN. (h) $U_{\text{Hfe}}\text{SRVAE}^3_{lc}$. (i) $U_{\text{Hfe}}\text{SRVAE}^3_{sq}$. (j) $U_{\text{Hfe}}\text{SRVAE}_3$.

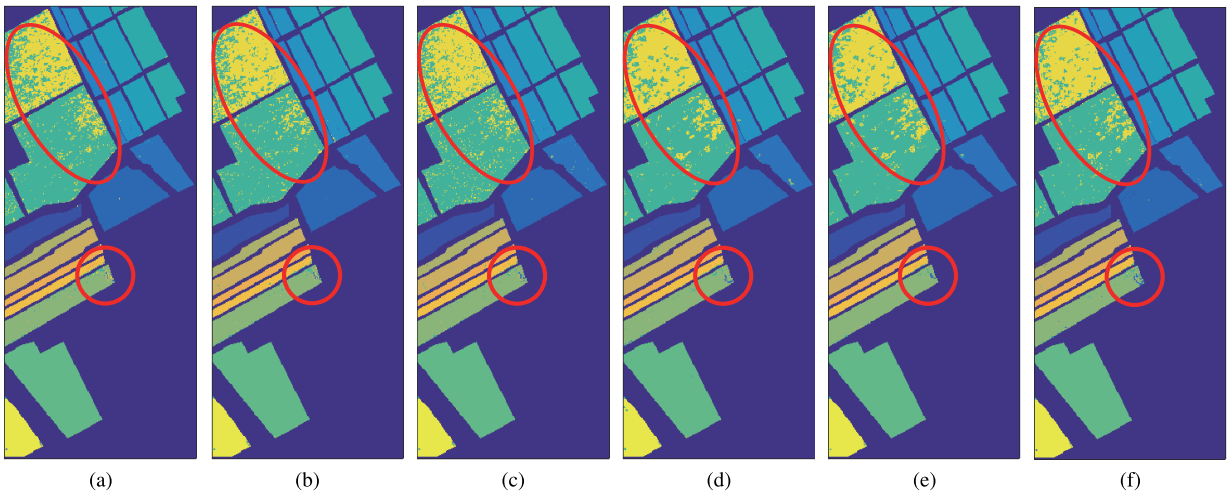


Fig. 12. Classification maps of the Salinas data set obtained by the comparison methods. (a) AE. (b) S-SAE. (c) VAE. (d) SAELR. (e) S-CNN. (f) $U_{\text{Hfe}}\text{SRVAE}$.

those of $U_{\text{Hfe}}\text{SRVAE}$ (such as Label 9 in LLE and Label 4 in SAELR), the proposed method yields the best classification results according to OAs, AAs, and κ values. $U_{\text{Hfe}}\text{SRVAE}^3_{lc}$ and $U_{\text{Hfe}}\text{SRVAE}^3_{sq}$ are methods using only one kind of spatial features, which correspond to $U_{\text{Hfe}}\text{SRVAE}_3$. Results show that using both kinds of spatial features could raise classification accuracies. Classification maps that correspond to Table IX are shown in Fig. 11. Some typical classes and parts are circled by red lines, and the results in the three upper regions illustrate that $U_{\text{Hfe}}\text{SRVAE}$ obtains a better map than other methods. Samples in class 2 (meadows) using all methods in lower square region share similar distinguish

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

TABLE X
CLASSIFICATION ACCURACIES (%) OF COMPARISON METHODS FOR THE SALINAS DATA SET, BOLD VALUES INDICATE THE BEST RESULTS FOR EACH ROW

	Origin	AE	SSAE	VAE	SAELR	SCNN	proposed
1	99.56	99.28	99.67	**99.78**	99.34	99.45	99.50
2	99.88	99.91	99.97	99.97	**100.00**	**100.00**	99.64
3	96.91	99.27	**99.49**	98.20	98.37	99.33	99.27
4	98.96	98.96	99.28	99.12	**99.68**	99.44	99.52
5	99.26	98.76	98.22	99.05	99.46	**99.71**	98.96
6	99.47	99.64	99.61	99.78	99.80	**99.94**	99.72
7	99.57	99.63	99.69	99.60	99.57	99.63	**99.72**
8	88.86	88.41	90.47	88.52	90.50	90.17	**90.56**
9	99.27	99.82	99.82	99.70	99.71	**99.86**	99.82
10	94.47	92.81	94.24	94.44	94.27	**97.08**	93.97
11	**98.34**	97.81	96.98	98.02	95.84	97.71	97.81
12	98.50	99.37	99.65	98.90	**99.94**	99.77	99.65
13	97.69	98.91	**99.03**	98.91	97.82	97.69	98.67
14	92.94	92.21	95.95	95.33	96.16	95.74	**97.40**
15	72.31	64.49	69.21	70.88	77.05	76.95	**79.18**
16	**98.59**	98.03	98.09	98.03	97.48	98.34	98.22
OA	92.91	91.86	93.09	92.88	94.12	94.33	**94.49**
AA	95.85	95.46	96.21	96.14	96.56	96.93	**96.98**
κ	92.10	90.92	92.29	92.06	93.44	93.68	**93.86**

abilities, which proves that these methods work well for samples in class 2.

3) Salinas Data Set: The Salinas data set is also used to further illustrate the effectiveness of the proposed U_{Hfe}SRVAE. Parameters used are the same as those for the first two data sets. Table X shows the classification accuracies obtained when the latent dimension is equal to 40, and all 204 bands are utilized when the original data set is directly used. The proposed U_{Hfe}SRVAE works better than the other comparison methods on the whole. Compared with SAELR and S-CNN, U_{Hfe}SRVAE increases OA by 0.37% and 0.16%. Although the promotion is not very high, the superiority of the proposed method should not be ignored. Fig. 12 shows the classification maps on the Salinas data set that correspond to Table X. Upper elliptic region shows that the proposed method yields better appearance, and samples using all methods in lower circular region share similar distinguish abilities. On the whole, the classification map obtained by U_{Hfe}SRVAE is closer to the ground truth.

V. CONCLUSION

In this article, we have proposed a novel unsupervised hyperspectral feature extraction architecture called U_{Hfe}SRVAE. Considering that positions of pixels in hyperspectral images are interrelated in the spatial domain, their neighbor regions can be seen as sequential data. Spatial features based on local sensing and sequential sensing are extracted by various network architectures in different aspects, including multilayer CNNs and long short-term memory networks. Moreover, spectral features are extracted by multilayer encoder, and the latent space vectors are generated from obtained means and standard deviations. Since the spatial features and the spectral features are both latent space features of the hyperspectral image, the former can be used to revise the latter. The proposed network architecture is designed for this purpose, and a novel loss function is also introduced for this network to guarantee the consistency of the probability distributions of various latent spatial features, which obtained from the same neighbor region. Experiments are conducted on three publicly available hyperspectral data sets, and then, several DNN techniques are chosen as comparison methods. Hyperparameters are determined by multiple experiments. Experimental results show that U_{Hfe}SRVAE achieves better classification results. The classification map obtained by the proposed method is more accurate than others. The performance of U_{Hfe}SRVAE in three-view drawing also shows its great clusterability and discernibility.

In the future, further experiments will focus on feature extraction generative models. Based on the spatial revising strategy, VAE achieves great results for classification. Therefore, we hope that this model can play a great role in other domains, including hyperspectral regression and target tracking.

REFERENCES

[1] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[2] H. Bi, J. Sun, and Z. Xu, "A graph-based semisupervised deep learning model for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2116–2132, Apr. 2019.

[3] M. E. Paoletti *et al.*, "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.

[4] Y. Liu, J. Yao, X. Lu, M. Xia, X. Wang, and Y. Liu, "RoadNet: Learning to comprehensively analyze road networks in complex urban scenes from high-resolution remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2043–2056, Apr. 2019.

[5] Y. Dong, B. Du, L. Zhang, and L. Zhang, "Dimensionality reduction and classification of hyperspectral images using ensemble discriminative local metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2509–2524, May 2017.

[6] L. Zhang, Y. Zhong, B. Huang, J. Gong, and P. Li, "Dimensionality reduction based on clonal selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4172–4186, Dec. 2007.

[7] J. An, X. Zhang, H. Zhou, and L. Jiao, "Tensor-based low-rank graph with multimanifold regularization for dimensionality reduction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4731–4746, Aug. 2018.

[8] Y.-J. Deng, H.-C. Li, K. Fu, Q. Du, and W. J. Emery, "Tensor low-rank discriminant embedding for hyperspectral image dimensionality reduction," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7183–7194, Dec. 2018.

[9] S. Zhang, "Studies of high spectral resolution atmospheric sounding data compression and noise reduction based on principal component analysis method," in *Proc. 2nd Int. Congr. Image Signal Process.*, Oct. 2009, pp. 1–5.

[10] P. N. Belhumeur, J. P. Hespanha, and D. J. Kriegman, "Eigenfaces vs. Fisherfaces: Recognition using class specific linear projection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 19, no. 7, pp. 711–720, Jul. 1997.

[11] C. Zhao, Y. Wang, and F. Mei, "Kernel ICA feature extraction for anomaly detection in hyperspectral imagery," *Chin. J. Electron.*, vol. 21, no. 2, pp. 265–269, 2012.

[12] M. Belkin and P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Comput.*, vol. 15, no. 6, pp. 1373–1396, Jun. 2003.

[13] S. T. Roweis, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[14] Z. Zhang and H. Zha, "Nonlinear dimension reduction via local tangent space alignment," in *Proc. Intell. Data Eng. Automated Learn. (IDEAL)*, Hong Kong, vol. 2690, Mar. 2003, pp. 477–481.

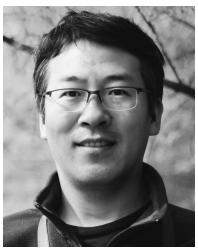
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.

[15] C. Deng, Y. Xue, X. Liu, C. Li, and D. Tao, "Active transfer learning network: A unified deep joint spectral–spatial feature learning model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1741–1754, Mar. 2019.
[16] J. Li, X. Huang, and J. Gong, "Deep neural network for remote-sensing image interpretation: Status and perspectives," *Nat. Sci. Review*, vol. 6, pp. 1082–1086, May 2019.
[17] N. He *et al.*, "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.
[18] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.
[19] X. Zhou *et al.*, "A deep manifold learning approach for spatial-spectral classification with limited labeled training samples," *Neurocomputing*, vol. 331, pp. 138–149, Feb. 2019.
[20] P. V. Arun, K. M. Buddhiraju, and A. Porwal, "CNN based sub-pixel mapping for hyperspectral images," *Neurocomputing*, vol. 311, pp. 51–64, Oct. 2018.
[21] Q. Zhang and L. Zhang, "Convolutional adaptive denoising autoencoders for hierarchical feature extraction," *Frontiers Comput. Sci.*, vol. 12, no. 6, pp. 1140–1148, Dec. 2018.
[22] M. Ning, P. Yu, W. Shaojun, and G. Wei, "A weight SAE based hyperspectral image anomaly targets detection," in *Proc. 13th IEEE Int. Conf. Electron. Meas. Instrum. (ICEMI)*, Oct. 2017, pp. 511–515.
[23] X. Han, Y. Zhong, and L. Zhang, "Spatial-spectral unsupervised convolutional sparse auto-encoder classifier for hyperspectral imagery," *Photogrammetric Eng. Remote Sens.*, vol. 83, no. 3, pp. 195–206, Mar. 2017.
[24] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. Int. Conf. Learn. Represent.*, 2014, pp. 1–14.
[25] N. Alami, M. Meknassi, and N. En-nahnahi, "Enhancing unsupervised neural networks based text summarization with word embedding and ensemble learning," *Expert Syst. Appl.*, vol. 123, pp. 195–211, Jun. 2019.
[26] C. Wu, F. Wu, S. Wu, Z. Yuan, J. Liu, and Y. Huang, "Semi-supervised dimensional sentiment analysis with variational autoencoder," *Knowl.-Based Syst.*, vol. 165, pp. 30–39, Feb. 2019.
[27] C. Shi and C.-M. Pun, "Multi-scale hierarchical recurrent neural networks for hyperspectral image classification," *Neurocomputing*, vol. 294, pp. 82–93, Jun. 2018.
[28] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.
[29] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
[30] A. Graves, *Supervised Sequence Labelling With Recurrent Neural Networks* (Studies in Computational Intelligence), vol. 385. Berlin, Germany: Springer-Verlag, 2012.
[31] Y. Guo, S. Han, H. Cao, Y. Zhang, and Q. Wang, "Guided filter based deep recurrent neural networks for hyperspectral image classification," *Procedia Comput. Sci.*, vol. 129, pp. 219–223, Jan. 2018.
[32] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Hyperspectral image classification using spectral-spatial LSTMs," *Neurocomputing*, vol. 328, pp. 39–47, Feb. 2019.
[33] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.
[34] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.
[35] X. Huang, X. Han, S. Ma, T. Lin, and J. Gong, "Monitoring ecosystem service change in the city of shenzhen by the use of high-resolution remotely sensed imagery and deep learning," *Land Degradation Develop.*, vol. 30, no. 12, pp. 1490–1501, Jul. 2019.
[36] T. Ma, J. Chen, and C. Xiao, "Constrained generation of semantically valid graphs via regularizing variational autoencoders," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 7113–7124.
[37] Y. Li, Y. Zhang, X. Huang, H. Zhu, and J. Ma, "Large-scale remote sensing image retrieval by deep hashing neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 950–965, Feb. 2018.
[38] J. Zabalza *et al.*, "Novel segmented stacked autoencoder for effective dimensionality reduction and feature extraction in hyperspectral imaging," *Neurocomputing*, vol. 185, pp. 1–10, Apr. 2016.
[39] M. Ahmad, M. A. Alqarni, A. M. Khan, R. Hussain, M. Mazzara, and S. Distefano, "Segmented and non-segmented stacked denoising autoencoder for hyperspectral band reduction," *Optik*, vol. 180, pp. 370–378, Feb. 2019.
[40] J. Pei, Y. Huang, W. Huo, Y. Zhang, J. Yang, and T.-S. Yeo, "SAR automatic target recognition based on multiview deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2196–2210, Apr. 2018.
[41] Q. Zhang, Q. Yuan, C. Zeng, X. Li, and Y. Wei, "Missing data reconstruction in remote sensing image with a unified spatial–temporal–spectral deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4274–4288, Aug. 2018.
[42] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.
[43] A. Belwalkar, A. Nath, and O. Dikshit, "Spectral-spatial classification of hyperspectral remote sensing images using variational autoencoder and convolution neural network," *ISPRS-Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, no. 5, pp. 613–620, Nov. 2018.
[44] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.
[45] F. Luo, L. Zhang, B. Du, L. Zhang, and Y. Dong, "Discriminant spatial-spectral hypergraph learning for hyperspectral image classification," in *Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 8480–8483.
[46] D. P. Kingma and J. L. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, May 2015, pp. 1–15.
[47] J. Duchi, E. Hazan, and Y. Singer, "Adaptive subgradient methods for online learning and stochastic optimization," *J. Mach. Learn. Res.*, vol. 12, pp. 2121–2159, Feb. 2011.
[48] T. Tieleman and G. Hinton, "Lecture 6.5-RMSProp: Divide the gradient by a running average of its recent magnitude," COURSERA: Neural Netw. Mach. Learn., Toronto, ON, Canada, Tech. Rep., 2012, vol. 4, pp. 26–30.



Wenbo Yu (Member, IEEE) received the B.S. degree in control science and engineering from the Harbin Institute of Technology, Harbin, China, in 2016, where he is pursuing the Ph.D. degree in control science and engineering.

His research interests include pattern recognition, signal processing and deep learning, and hyperspectral image analysis.



Miao Zhang (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in control science and engineering from the Harbin Institute of Technology, Harbin, China, in 2002, 2004, and 2009, respectively.

He is an Associate Professor with the School of Astronautics, Harbin Institute of Technology. His research interests include signal and image processing, fault diagnosis of control systems, and embedded systems.



Yi Shen (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in control science and engineering from the Harbin Institute of Technology, Harbin, China, in 1985, 1988, and 1995, respectively.

He is a Professor with the School of Astronautics, Harbin Institute of Technology. His research interests include instrumentation and measurement, ultrasound signal processing, and modern detection techniques.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:29:47 UTC from IEEE Xplore. Restrictions apply.